



Fieldnote

Entries made in the moment, even offline.
No confirmed entry is lost.

THE PROBLEM, IN PAPER DIARIES

11% actual versus 90% reported

Participant

Android app

- ✓ Joins by QR, code or invitation
- ✓ Text, photo, video, audio, scale
- ✓ Records offline, no duplicates
- ✓ Optional account, deletable data

Researcher

web portal

- ✓ Creates studies, activities and rules
- ✓ Invites: code, poster or pool
- ✓ Adherence and entries in real time
- ✓ Recorded time and received time
- ✓ Tags and exports the data

Administrator

web portal

- ✓ Researcher accounts with TOTP
- ✓ All studies and the pool
- ✓ Tamper-proof audit log
- ✓ Deletion requests

Security and privacy

pseudonyms (P-07)

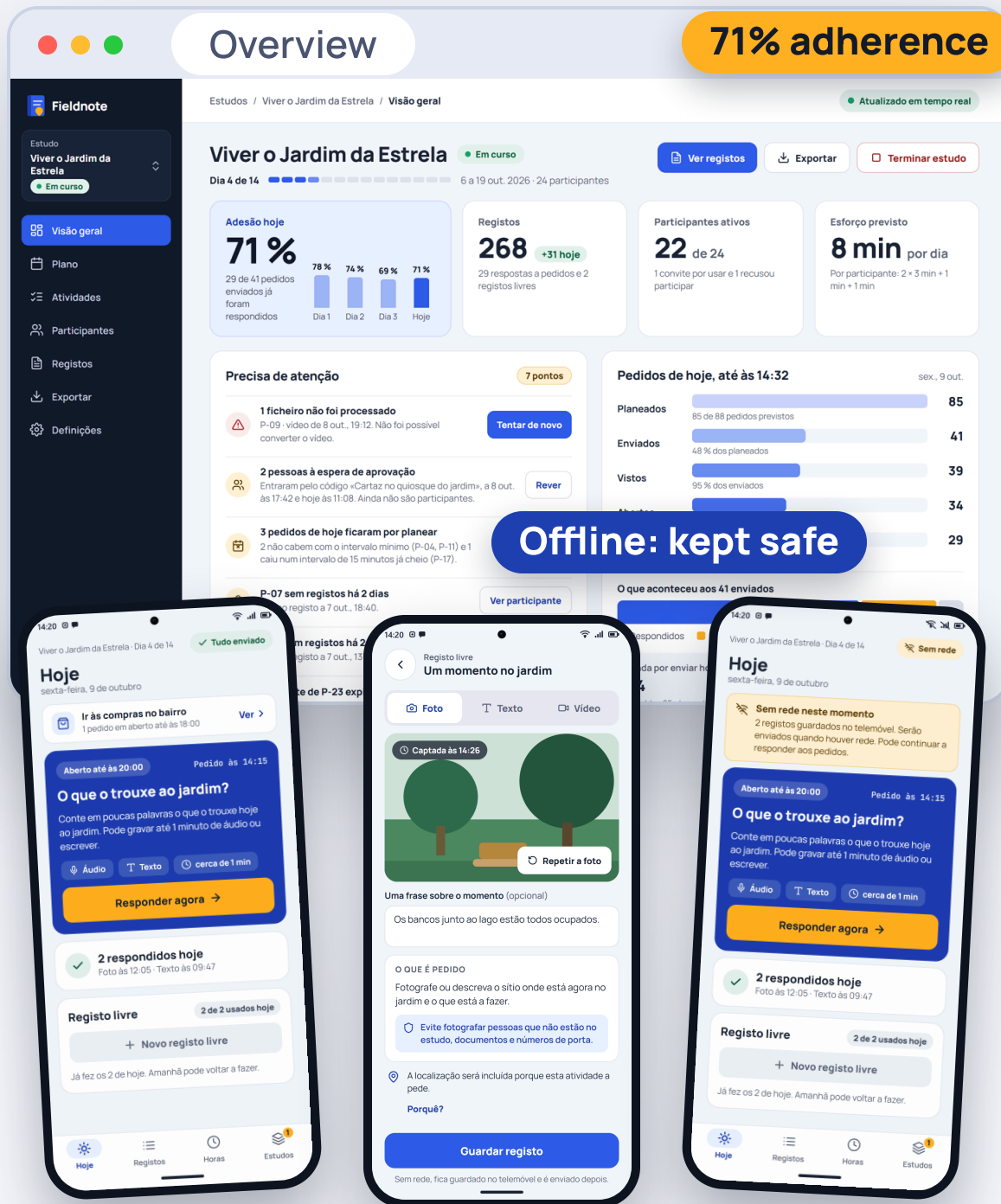
roles

signed URLs

EXIF removed

audit

delete on day 7



71% adherence

Offline: kept safe

Architecture

Tailscale · 2 Traefik gateways

API

PostgreSQL

RabbitMQ

Garage (S3)

Patroni, etcd and quorum queues. Each file has 3 copies. Keeps receiving if a node goes down.

Scheduler (AI)

Deciding when to ask for each activity is a constraint satisfaction problem (CSP).

availability

activity window

minimum gap

daily limit

max. per 15 min

Backtracking, MRV, LCV, forward checking and AC-3. Approved by the AI lecturer.

Validated stack by PDS and SD

App React Native · Expo · TypeScript · SQLite · Expo Push

Web React · Vite · Python 3.12 · FastAPI · ffmpeg

Data PostgreSQL 17 · Patroni · etcd · RabbitMQ · Garage

Infra Compose · Tailscale · Traefik · Prometheus · Grafana

2026 calendar

7 Oct

pitch

9 Oct

1st delivery

13 Nov

prototype

18 Dec

final delivery